

No. of Functional Features Included in the Solution

Date	05 July 2026	Team ID	—
Project Name	India's Agriculture Crop Production Analysis	Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Data Collection & Integration	Collects crop production data from multiple sources including government portals, APIs, and datasets.	Data Layer	Done	
2	Data Cleaning & Preprocessing	Cleans and preprocesses raw data by handling missing values, duplicates, and inconsistencies.	Data Layer	Done	
3	Exploratory Data Analysis (EDA)	Performs statistical analysis to identify trends in crop production by state, season, and year.	Analytics Module	Done	
4	Crop Production Analysis	Analyzes top-producing crops, productivity patterns, and growth trends over time.	Analytics Module	Done	
5	Interactive Dashboard	Provides interactive visualizations (charts, graphs, maps) to explore crop production insights.	Visualization Module	Done	
6	State & Crop Comparison	Allows users to compare crop production across states, seasons, and years.	Analytics / Visualization	Done	
7	Report Generation & Export	Generates summary reports and exports data/insights in CSV, PDF, and Excel formats.	Reporting Module	Done	
8	User Management & Security	Handles user authentication and role-based access for secure data and dashboard access.	Security / Database	Done	

Feature Summary:

INDIA'S AGRICULTURE CROP PRODUCTION ANALYSIS

Feature Summary & Category Breakdown

Metric	Count / Value
Total Features Planned	10
Total Features Implemented	10
Core / Must-Have Features	7
Additional / Nice-to-Have Features	3
Features Tested & Verified	10

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	 User Interface (UI)	2	Interactive Dashboard, GIS Maps & Charts Display, Filter & Search Interface
2	 Backend / Logic	3	Data Processing Pipeline, Analytics Engine (ML Models), Report Generation Service
3	 Database / Storage	2	Crop Production Database, Soil & Weather Data Storage
4	 API / Integration	1	Weather API Integration, Government APIs (eNAM, AgriStack)
5	 Security / Authentication	1	Secure User Login, Role-Based Access Control
6	 Notifications / Alerts	1	Weather Alerts, Pest & Disease Notifications